9 Find the area under the curve, to four decimal places, for each part of the standardised normal curves described below:

- **a** To the left of $z = 1.45$
- **b** To the right of $z = 1.58$
- **c** To the left of $z = -1.23$
- **d** To the right of $z = -1.17$
- **e** Between $z = 1.52$ and $z = 1.87$
- **f** Between 1.10 and 1.60 standard deviations above the mean.

10 Calculate the percentage of standardised data, to two decimal places, that is:

- **a** Greater than $z = -1.51$
- **b** Between $z = -1.14$ and $z = 2.37$

11 Calculate the probability, to four decimal places, that a z-score is:

- **a** Either at most -1.08 or greater than 2.07
- **b** Greater than -0.63 and at most 1.44
- **c** At most 1.60 given that it is greater than -0.69
- **d** At most 1.03 given that it is less than 2.58

Calculations with the standard and general normal distributions

12 Using your calculator, find the value of k to four decimal places for the following probabilities:

- **a** The probability of a z-score being at most k is equal to 0.8031 in the standard normal distribution.
- **b** The probability of a z-score being greater than k is equal to 0.7934 in the standard normal distribution.
- **c** The probability of a z-score being at most k is equal to 0.218 in the standard normal distribution.
- **d** The probability of a z-score being greater than k is equal to 0.1562 in the standard normal distribution.
- **e** The probability of a z-score being greater than $-k$ and at most k is equal to 0.6123 in the standard normal distribution.
- **f** The probability of a z-score being greater than -2.11 and at most k is equal to 0.8273 in the standard normal distribution.

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13 If $X \sim N\left(20, 5^2\right)$, use your calculator to find the value of k in the following parts:

a $P(X < k) = 0.65$

b $P(X > k) = 0.45$

c $P(k < X < 27) = 0.89$

d $P(21 < X < k) = 0.4$

14 Languages and Mathematics are very different disciplines, and so to compare results in the two subjects, the standard deviation is used. The mean and standard deviation of exam results in each subject are given in the table:

	Mean	Std. Deviation
Languages	60	7
Mathematics	67	8

a A student receives a mark of 81 in Languages. How many standard deviations away from the mean is this mark?

b What mark in Mathematics would be equivalent to a mark of 81 in Languages?

c A student receives a mark of 86.2 in Mathematics. How many standard deviations away from the mean is this mark? Round your answer to one decimal place.

d What mark in Languages would be equivalent to a mark of 86.2 in Mathematics? Round your answer to one decimal place.

Quantiles and percentiles

15 For the standard normal variable $X \sim N(0, 1)$, use a CAS to determine the following values to three decimal places:

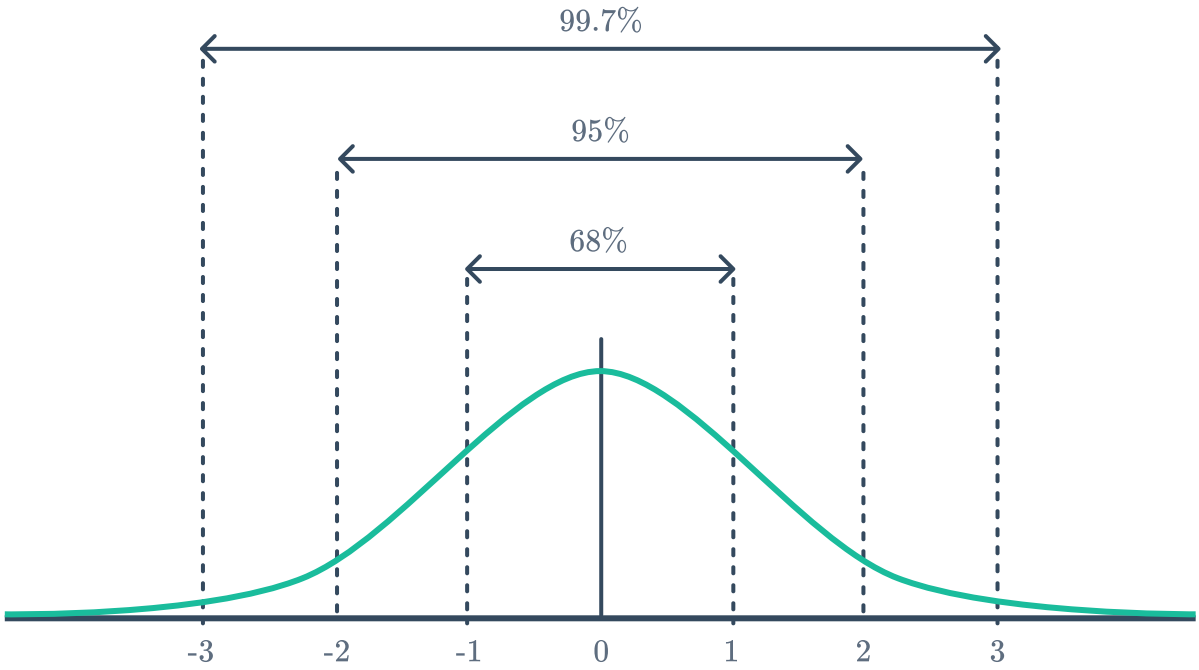
a The 0.7 quantile

b The 65th percentile

c The lowest score in the top 20 percent

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16 Consider the graph of a standard normal distribution showing the $68-95-99.7$ rule:



a Which value is the closest to the 0.5 quantile?

b Which value is the closest to the 0.84 quantile?

c Which value is the closest to the 16th percentile?

17 Consider a normal distribution defined by $X \sim N(50, 25)$. Use the $68-95-99.7$ rule to answer the following questions:

a Which value is equivalent to the 0.16 quantile?

b Which value is equivalent to the 0.025 quantile?

c Which value is equivalent to the 97.5th percentile?

18 The heights of a certain species of fully grown plants are thought to be normally distributed with a mean of 40 cm and a standard deviation of 1 cm. Use the $68-95-99.7$ rule to answer the following questions:

a What is the height of the shortest plant in the 84th percentile?

b What is the height of the shortest plant in the 0.0015 quantile?

19 For a normal variable defined by $X \sim N(100, 100)$, use a CAS to determine the following values to three decimal places:

a The 0.2 quantile

b The 90th percentile

c The lowest score that is greater than the bottom 30 percent

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20 If $X \sim N\left(30, 4^2\right)$, calculate:

a The 0.5 quantile

b The 0.83 quantile

c The 35th percentile

21 A random variable is normally distributed such that $X \sim N\left(50, 25\right)$.

a Calculate the standard score if $X = 58$.

b Using a CAS calculator or otherwise, calculate the z-score for the 0.35 quantile.

c Hence, find the X value for the 0.35 quantile.

22 Mensa is an organisation that only accepts members who score in the 98th percentile or above in an IQ test. Explain what a person has to do to get into Mensa.

Unknown mean or standard deviation

23 If Maximilian scores 57.6, with a z score of 2, in a test that has a standard deviation of 5.8, what was the mean score?

24 If Han scores 43.2, with a z score of -3, in a test that has a standard deviation of 5.6, what was the mean score?

25 If Luke scores 68, for a z score of -3, in a test that has a mean score of 93.5, what was the standard deviation of the test scores?

26 If Saoirse scores 32.5, for a z score of -4, in a test that has a mean score of 58.5, what was the standard deviation of the test scores?

27 If $X \sim N\left(\mu, 100\right)$, use your calculator to find μ if $P\left(\mu \leq X \leq 20\right) = 0.3013$. Round your answer to two decimal places.

28 If $X \sim N\left(\mu, 100\right)$, use your calculator to find μ if $P\left(\mu \leq X \leq 30\right) = 0.419$. Round your answer to two decimal places.

29 If $X \sim N\left(\mu, \sigma^2\right)$, use your calculator to find μ and σ if $P\left(X < 70\right) = 0.1817$ and $P\left(X < 80\right) = 0.9655$. Round your answers to two decimal places.

30 If $X \sim N\left(\mu, \sigma^2\right)$, use your calculator to find μ and σ if $P\left(X < 12\right) = 0.2859$ and $P\left(X < 18\right) = 0.8677$. Round your answer to two decimal places.

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